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VEHICLE DISPLAY DEVICE AND IMAGE DISPLAY METHOD

Abstract

A vehicle display device includes: an image acquisition unit that acquires a peripheral image being an image obtained by imaging a peripheral region of a vehicle, the peripheral region including a rear region of the vehicle; and a display device that displays the peripheral image in a display region in a windshield of the vehicle, and the display region is an upper region in the windshield, and is a region that is continuous from one end to another end in a vehicle width direction in the windshield.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation application of International Application No. PCT/JP2023/044599 filed on Dec. 13, 2023 which claims the benefit of priority from Japanese Patent Application No. 2022-209268 filed on Dec. 27, 2022 and designating the U.S., the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to a vehicle display device and an image display method.

BACKGROUND

2. Description of the Related Art

[0003] Conventionally, a technique for displaying an image on a windshield has been proposed. JP 2020-011 586 A discloses a vehicle windshield structure that enables a video indicating a rear side of a vehicle to be visually recognized with satisfaction even in a case where the periphery of the vehicle is bright. The vehicle windshield structure of JP 2020-011 586 A is provided on a vehicle upper side in a windshield, and includes a display unit that is transparent and on which the video indicating the rear side of the vehicle is displayed, and a light shielding member that shields, from light, a region including the display unit in the windshield in a case where the periphery of the vehicle has brightness that is greater than predetermined brightness.

[0004] It is desirable that an amount of movement of a line of sight can be reduced in a case where a driver checks the periphery of the vehicle including the rear side of the vehicle. For example, in a case where a mirror that indicates the rear side of the vehicle is located apart from a mirror that indicates a diagonally rear side of the vehicle, a line of sight greatly moves to check the periphery.

SUMMARY OF THE INVENTION

[0005] It is an object of the present invention to provide a vehicle display device and an image display method that can reduce an amount of movement of a line of sight of a driver in a case where the driver checks the periphery of a vehicle including a rear side of the vehicle.

[0006] In order to achieve the above mentioned object, a vehicle display device according to one aspect of the present invention includes an image acquisition unit that acquires a peripheral image being an image obtained by imaging a peripheral region of a vehicle, the peripheral region including a rear region of the vehicle; a display device that displays the peripheral image in a display region in a windshield of the vehicle; and a generation unit that generates a caution image being an image for cautioning against a target to be careful about that is included in the peripheral image, wherein the display region is an upper region in the windshield, and is a region that is continuous from one end to another end in a vehicle width direction in the windshield, the target to be careful about at the time of backward movement of the vehicle is an object that is present on a route behind the vehicle or an object that is present near the route, the target to be careful about at the time of a forward movement of the vehicle is an object that is approaching to the vehicle from a rear side, and the display device displays the peripheral image and the caution image in the display region.

[0007] The above and other objects, features, advantages and technical and industrial significance of this invention will be better understood by reading the following detailed description of presently preferred embodiments of the invention, when considered in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a diagram illustrating a display device according to an embodiment;
[0009] FIG. 2 is a sectional view illustrating the display device according to the embodiment;
[0010] FIG. 3 is a diagram illustrating a display region according to the embodiment;
[0011] FIG. 4 is a block diagram of a vehicle display device according to the embodiment;
[0012] FIG. 5 is a diagram explaining an imaging range according to the embodiment;
[0013] FIG. 6 is a diagram illustrating an example of a peripheral image according to the embodiment;
[0014] FIG. 7 is a diagram illustrating an example of the peripheral image according to the embodiment;
[0015] FIG. 8 is a flowchart relating to an image display method according to the embodiment;
[0016] FIG. 9 is a diagram illustrating the peripheral image and a caution image according to the embodiment;
[0017] FIG. 10 is a diagram illustrating an example of a camera according to the embodiment; and
[0018] FIG. 11 is a diagram illustrating a schematic configuration of a display device according to a variation of the embodiment.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0019] A vehicle display device and an image display method according to an embodiment of the present invention will be described in detail below with reference to the drawings. Note that this embodiment is not restrictive of this invention. Furthermore, components in the embodiment described below include components that those skilled in the art could easily conceive of, or substantially the same components.

EMBODIMENT

[0020] An embodiment will be described with reference to FIGS. 1 to 10. The present embodiment relates to a vehicle display device and an image display method. FIG. 1 is a diagram illustrating a display device according to the embodiment, FIG. 2 is a sectional view illustrating the display device according to the embodiment, FIG. 3 is a diagram illustrating a display region according to the embodiment, FIG. 4 is a block diagram of a vehicle display device according to the embodiment, FIG. 5 is a diagram explaining an imaging range according to the embodiment, FIGS. 6 and 7 are diagrams illustrating an example of a peripheral image according to the embodiment, FIG. 8 is a flowchart relating to an image display method according to the embodiment, FIG. 9 is a diagram illustrating the peripheral image and a caution image according to the embodiment, and FIG. 10 is a diagram illustrating an example of a camera according to the embodiment. FIG. 2 illustrates cross section II-II of FIG. 1.

[0021] FIG. 1 illustrates a display device 2 according to the present embodiment. The display device 2 according to the present embodiment is a display device that has a transparent film shape and is disposed in a windshield 110 of a vehicle 100. The display device 2 is a light-emitting display, and is, for example, an organic EL display (OLED). The illustrated display device 2 is disposed over the entire face of the windshield 110. The display device 2 is disposed, for example, on an inner face 110a in the windshield 110, as illustrated in FIG. 2. The inner face 110a is a face on a rear side X2 in a vehicle forward/backward direction X in the windshield 110, and is a face that faces a driver 200. Note that a position of the display device 2 is not limited to the inner face 110a. The display device 2 may be disposed, for example, in a laminated glass included in the windshield 110.

[0022] As illustrated in FIG. 2, an edge of the windshield 110 according to the present embodiment adheres to a vehicle body 170 of the vehicle 100, and is held by the vehicle body 170. The display device 2 is disposed to cover the windshield 110 from one end 110b to another end 110c in a

vehicle width direction Y. Here, the one end **110b** and the other end **110c** of the windshield **110** are boundaries of a region that is exposed to a vehicle interior side. Stated another way, the display device **2** covers the entirety from a left-hand end to a right-hand end of a portion that is visually recognized by the driver **200** in the windshield **110**. Similarly, the display device **2** covers the entirety from an upper end to a lower end of the windshield **110**. Stated another way, the display device **2** covers the entire face of the region that is visually recognized by the driver **200** in the windshield **110**. Note that, for the reasons of mounting on the vehicle **100**, the display device **2** may be disposed in such a way that the one end **110b** and the other end **110c** are exposed. Stated another way, there may be a slight clearance between the vehicle body **170** and the display device **2**.

[0023] A vehicle display device **1** according to the present embodiment causes the display device **2** to display an image in a display region **120** of the windshield **110**, as illustrated in FIG. **3**. The display region **120** is an upper region in the windshield **110**, and is a region that is continuous from one end to another end in the vehicle width direction Y in the windshield **110**. It is preferable that one end **120b** of the display region **120** match the one end **110b** of the windshield **110**. It is preferable that another end **120c** of the display region **120** match the other end **110c** of the windshield **110**. The illustrated display region **120** is a single region that is continuous from the one end **110b** to the other end **110c** of the windshield **110**. A shape of the display region **120** is a belt shape that extends in the vehicle width direction Y.

[0024] A specific shape of the display region **120** is determined in accordance with a shape of the windshield **110**. The shape of the display region **120** is, for example, a trapezoid or a rectangle. In a case where an upper side in the windshield **110** is curved, the display region **120** may be curved to correspond to this side.

[0025] FIG. **4** is a block diagram of the vehicle display device **1** according to the present embodiment. The vehicle display device **1** includes the display device **2** and a control unit **3**. The control unit **3** is a display control unit that includes a computer including an arithmetic unit and a memory, and an input/output interface. The control unit **3** may be configured as an electronic control unit (ECU). In the memory of the control unit **3**, a program or data for performing operations according to the present embodiment has been stored in advance.

[0026] The control unit **3** includes an image acquisition unit **31**, an information acquisition unit **32**, a detection unit **33**, a generation unit **34**, and an output unit **35**. The image acquisition unit **31**, the information acquisition unit **32**, the detection unit **33**, the generation unit **34**, and the output unit **35** may be functions that are achieved according to the program, or may be dedicated processing circuits.

[0027] The image acquisition unit **31** acquires a peripheral image **40** from a camera **130** equipped in the vehicle **100**. The peripheral image **40** is an image obtained by imaging a peripheral region of the vehicle **100** including a rear region of the vehicle **100**. FIG. **5** illustrates an example of disposition of the camera **130**. The camera **130** illustrated in FIG. **5** is disposed in a rear portion of the vehicle **100**, and images the peripheral region of the vehicle **100**. More specifically, the camera **130** images a rear region Ra of the vehicle **100**, a diagonally left rear region Rb of the vehicle **100**, and a diagonally right rear region Rc of the vehicle **100**.

[0028] An imaging range of the camera **130** is determined on the basis of a rear viewing angle θ illustrated in FIG. **5**. The rear viewing angle θ is an angle of a pseudo field of view in a case where a rear side is viewed from a predetermined point P1 of the vehicle **100**, and is an angle in a horizontal direction. Stated another way, the rear viewing angle θ corresponds to an angle of view in the horizontal direction of the camera **130** in a case where the camera **130** has been installed at the predetermined point P1. The camera **130** is configured to image at least a region of the rear viewing angle θ in a case where the rear side of the vehicle is viewed from the predetermined point P1 to be able to generate an image indicating the region of the rear viewing angle θ . The predetermined point P1 is, for example, an eye point EP of the vehicle **100**. The predetermined point P1 may be a center point of an eye range or a center point of an eye-box of the vehicle **100**.

The predetermined point P1 may be a point on the windshield **110**, and may be, for example, a center point in the vehicle width direction Y in the display region **120**.

[0029] FIG. **6** illustrates an example of the peripheral image **40** captured by the camera **130**. The peripheral image **40** includes a range **40a** in which the rear region Ra has been imaged, a range **40b** in which the diagonally left rear region Rb has been imaged, and a range **40c** in which the diagonally right rear region Rc has been imaged. Stated another way, the peripheral image **40** is a continuous image obtained by imaging a region from a diagonally left rear side to a diagonally right rear side of the vehicle **100**. The vehicle display device **1** causes the display device **2** to display the peripheral image **40** in the display region **120**.

[0030] FIG. **7** illustrates the peripheral image **40** that is displayed in the display region **120** of the windshield **110**. The driver **200** can check a situation in a wide region including a diagonally rear side of the vehicle **100** by using the peripheral image **40**. A region from the diagonally left rear side to the diagonally right rear side of the vehicle **100** is seamlessly displayed as a single image, and therefore the driver **200** can easily grasp a peripheral situation of the vehicle **100**.

[0031] In conventional general vehicles, the driver **200** needs to direct a line of sight to mirrors that are different depending on whether a rear side of the vehicle or a diagonally rear side of the vehicle will be checked. Accordingly, in a case where an attempt is made to check a wide range from a diagonally left rear side to a diagonally right rear side of the vehicle, a line of sight needs to greatly move. In contrast to this, the vehicle display device **1** according to the present embodiment enables a wide range to be checked by using a single peripheral image **40**. Therefore, the vehicle display device **1** according to the present embodiment can reduce an amount of movement of a line of sight of the driver **200** in checking the peripheral situation.

[0032] Furthermore, the vehicle display device **1** according to the present embodiment causes a caution image **50** to be displayed in a case where the peripheral image **40** includes a target to be careful about, as described below. As illustrated in FIG. **4**, the information acquisition unit **32** acquires various types of information from a sensor **140** and a vehicle ECU **150**. The sensor **140** is a sensor that is disposed in the vehicle **100**, and detects the peripheral situation of the vehicle **100**. The sensor **140** may include, for example, a LiDAR that detects a direction of and a distance to an object that is present in the periphery, may include a ranging sensor such as a millimeter wave radar, or may include another sensor or a stereoscopic camera.

[0033] The information acquisition unit **32** acquires, from the sensor **140**, information relating to a direction of an object that is present around the vehicle **100** and a distance from the vehicle **100** to the object. The information acquisition unit **32** may further acquire, from the sensor **140**, information relating to a shape of the object or information relating to the speed of movement of the object.

[0034] The vehicle ECU **150** is a control device that controls the vehicle **100**, and is, for example, an integrated control ECU that performs integrated control on the vehicle **100**. The information acquisition unit **32** acquires, from the vehicle ECU **150**, information of the vehicle **100**. The information acquisition unit **32** acquires, from the vehicle ECU **150**, information relating to, for example, the speed of traveling, acceleration, a shift position, a traveling mode, a turn signal, or a steering angle of the vehicle **100**. The information acquisition unit **32** may acquire information of route guidance provided by a navigation device from the vehicle ECU **150** or the navigation device.

[0035] Note that the information acquisition unit **32** may acquire a detection result of the sensor **140** via the vehicle ECU **150**. Furthermore, in a case where the vehicle ECU **150** has information relating to a direction of a peripheral object and a distance to the peripheral object, these pieces of information may be transmitted from the vehicle ECU **150** to the information acquisition unit **32**. The information acquisition unit **32** transmits, to the detection unit **33**, pieces of information that have been acquired from the sensor **140** and the vehicle ECU **150**.

[0036] The detection unit **33** detects the target to be careful about that is included in the peripheral image **40**. The detection unit **33** performs, for example, image recognition on the peripheral image

40 to extract an object indicated in the peripheral image **40**. The detection unit **33** determines whether the extracted object is a candidate object for the target to be careful about. The candidate object includes another vehicle, a moving body such as a pedestrian, or a structure such as a building.

[0037] The detection unit **33** determines an object that satisfies predetermined conditions as the target to be careful about from among the detected candidate objects. For example, in a case where the candidate object is another vehicle, the detection unit **33** determines whether the candidate object is the target to be careful about on the basis of the size of the candidate object. In this case, in a case where the size of the candidate object is greater than or equal to a predetermined value, the candidate object may be determined to be the target to be careful about. Similarly, in a case where the candidate object is a pedestrian and in a case where the size of the candidate object is greater than or equal to a predetermined value, the detection unit **33** may determine that the candidate object is the target to be careful about.

[0038] The detection unit **33** may determine the target to be careful about on the basis of information acquired from the information acquisition unit **32**. In this case, for example, information of a peripheral object that has been acquired from the sensor **140** is associated with the candidate object that has been extracted from the peripheral image **40**. The detection unit **33** determines whether the candidate object is the target to be careful about on the basis of sensor information that has been associated with the candidate object.

[0039] Here, an operation of the vehicle display device **1** according to the present embodiment will be described with reference to the flowchart of FIG. **8**. In Step **S10**, the image acquisition unit **31** acquires the peripheral image **40** from the camera **130**. When Step **S10** has been performed, the processing proceeds to Step **S20**.

[0040] In Step **S20**, the information acquisition unit **32** acquires pieces of information from the sensor **140** and the vehicle ECU **150**. When Step **S20** has been performed, the processing proceeds to Step **S30**.

[0041] In Step **S30**, the detection unit **33** determines whether a current shift position is an R position for backward movement. In a case where the determination of Step **S30** is affirmative to determine that the current shift position is the R position, the processing proceeds to Step **S40**, and in a case where the determination is negative, the processing proceeds to Step **S70**.

[0042] In Step **S40**, the detection unit **33** determines whether a target to be careful about at the time of backward movement has been detected. In a case where the shift position is the R position, an object that is present on a route behind the vehicle **100** or an object that is present near the route is determined as the target to be careful about. The detection unit **33** determines, for each of the extracted candidate objects, whether a candidate object is located on the route of the vehicle **100** or near the route. A candidate object that is located on the route or near the route is registered as the target to be careful about in a list.

[0043] The detection unit **33** may determine whether a candidate object is the target to be careful about on the basis of a distance from the vehicle **100** to the candidate object. For example, a candidate object for which a distance from the vehicle **100** is less than or equal to a lower limit value is determined as the target to be careful about. The detection unit **33** may determine whether a candidate object is the target to be careful about on the basis of the speed of the candidate object relative to the vehicle **100**. For example, a candidate object for which the speed of approach to the vehicle **100** is greater than or equal to an upper limit value is determined as the target to be careful about. In a case where the determination of Step **S40** is affirmative to determine that the target to be careful about has been detected, the processing proceeds to Step **S50**, and in a case where the determination is negative, the processing proceeds to Step **S100**.

[0044] In Step **S50**, the generation unit **34** generates the caution image **50** at the time of backward movement. The generation unit **34** generates the caution image **50** on the basis of a detection result of the detection unit **33**. The caution image **50** is an image for cautioning against the target to be

careful about. FIG. 9 illustrates an example of the caution image 50. The caution image 50 of FIG. 9 is an image that has been generated for a pedestrian that is located behind the vehicle 100. In other words, the caution image 50 of FIG. 9 is an image for cautioning the driver 200 against a pedestrian that is present on a rear side.

[0045] The illustrated caution image 50 is a rectangular image that surrounds a pedestrian serving as the target to be careful about. In FIG. 9, caution images 50 that are different in color in accordance with a distance to the target to be careful about are displayed. A caution image 50 of a color having a high degree of caution, for example, red, is displayed for a target to be careful about that is located near the vehicle 100. A caution image 50 of a color having a low degree of caution, for example, yellow green or yellow, is displayed for a target to be careful about that is a little far from the vehicle 100. The generation unit 34 generates the caution image 50 for the target to be careful about that has been detected in Step S40. When Step S50 has been performed, the processing proceeds to Step S60.

[0046] In Step S60, the output unit 35 causes the display device 2 to display the peripheral image 40 and the caution image 50. The output unit 35 outputs, to the display device 2, an image to be displayed. For example, the output unit 35 superimposes the caution image 50 acquired from the generation unit 34 onto the peripheral image 40 to generate a composite image. In this case, the output unit 35 outputs the composite image to the display device 2. The output unit 35 according to the present embodiment outputs the peripheral image 40 to the display device 2 without changing an aspect ratio of the peripheral image 40. Stated another way, an image captured by the camera 130 is displayed in the display device 2 in a state where an aspect ratio of the image is maintained. Note that the output unit 35 may correct the peripheral image 40 or the composite image in accordance with the shape of the windshield 110. For example, in a case where the shape of the display region 120 is a trapezoid, correction may be performed to process a rectangular image captured by the camera 130 into a trapezoid image. Note that the control unit 3 may cause a speaker 160 to output sound cautioning against the target to be careful about. When Step S60 has been performed, the flowchart terminates.

[0047] In Step S70, the detection unit 33 determines whether a target to be careful about at the time of a forward movement has been detected. In a case where the shift position is not different from the R position or in a case where the shift position is a position for a forward movement, the detection unit 33 detects the target to be careful about at the time of a forward movement. A target to be careful about in this case is an object that is approaching to the vehicle 100 from a rear side. For example, another vehicle that is approaching from a rear side on the same lane as a lane of the vehicle 100 is a candidate object. For example, in a case where the vehicle 100 changes lanes, another vehicle that is traveling on a lane of a movement destination is a candidate object. For example, a motorcycle that is about to pass the vehicle 100 is a candidate object.

[0048] From among these candidate objects, an object that satisfies predetermined conditions is determined as the target to be careful about. The predetermined conditions include at least one of, for example, a condition that a distance from the vehicle 100 to another vehicle is less than or equal to a lower limit value and a condition that the speed of approach of another vehicle to the vehicle 100 is greater than or equal to an upper limit value. In a case where the determination of Step S70 is affirmative to determine that the target to be careful about has been detected, the processing proceeds to Step S80, and in a case where the determination is negative, the processing proceeds to Step S100.

[0049] In Step S80, the generation unit 34 generates a caution image 50 at the time of a forward movement. The generation unit 34 generates the caution image 50 for the target to be careful about that has been detected in Step S70. The caution image 50 at the time of a forward movement is, for example, a frame-shaped image that surrounds another vehicle. When Step S80 has been performed, the processing proceeds to Step S90.

[0050] In Step S90, the output unit 35 causes the display device 2 to display the peripheral image

40 and the caution image **50**. The output unit **35** outputs, to the display device **2**, a composite image in which the caution image **50** has been superimposed onto the peripheral image **40**. When Step **S90** has been performed, the flowchart terminates.

[0051] In Step **S100**, the output unit **35** causes the display device **2** to display the peripheral image **40**. In this case, the peripheral image **40** that does not include the caution image **50** is displayed in the display region **120**, as illustrated in FIG. 7. When Step **S100** has been performed, the flowchart terminates.

[0052] The vehicle display device **1** according to the present embodiment enables the driver **200** to understand, at a glance, a direction in which the target to be careful about is present within a wide range behind the vehicle **100**.

[0053] The vehicle display device **1** according to the present embodiment displays a wide rear image in the display region **120**. The rear viewing angle θ illustrated in FIG. 5 is, for example, a value from 140° to 180° . It is preferable that the rear viewing angle θ be a value of 140° or more and 150° or less. The peripheral image **40** is captured at such a wide rear viewing angle θ , and this provides the driver **200** with information that is appropriate to check the periphery.

[0054] Note that the vehicle display device **1** may acquire the peripheral image **40** from a plurality of cameras **130**. The vehicle **100** illustrated in FIG. 10 is equipped with three cameras **130**. The cameras **130** include a first camera **130a**, a second camera **130b**, and a third camera **130c**. The first camera **130a** is disposed in a rear portion in the vehicle **100**, and images a rear side X2 of the vehicle **100**. The second camera **130b** is disposed in a left-hand side portion in the vehicle **100**. The second camera **130b** images a left-hand side and a diagonally left rear side of the vehicle **100**. The third camera **130c** is disposed in a right-hand side portion in the vehicle **100**. The third camera **130c** images a right-hand side and a diagonally right rear side of the vehicle **100**.

[0055] The second camera **130b** and the third camera **130c** are disposed on sides of the windshield **110**, and are disposed, for example, in positions that correspond to door mirrors. The second camera **130b** is disposed to be able to image an object Ob1 located on a left-hand side of the vehicle **100**. The third camera **130c** is disposed to be able to image an object Ob2 located on a right-hand side of the vehicle **100**.

[0056] The image acquisition unit **31** connects a plurality of images acquired from the first camera **130a**, the second camera **130b**, and the third camera **130c** to generate the peripheral image **40**. The peripheral image **40** that has been acquired as described above includes the objects Ob1 and Ob2 located on sides of the vehicle **100**. Therefore, the driver **200** can confirm the presence of the objects Ob1 and Ob2 located on sides, by using the peripheral image **40**.

[0057] In a case where a motorcycle that has approached to the vehicle **100** from a rear side passes on a left-hand side or a right-hand side of the vehicle **100**, the driver **200** can sequentially check a movement of the motorcycle, by using the peripheral image **40**. The driver **200** can view the peripheral image **40** by slightly moving a line of sight from a state of visually recognizing the forward. Therefore, the vehicle display device **1** according to the present embodiment makes it possible to check a rear side of the vehicle or a side of the vehicle with no need of a large movement of a line of sight. Furthermore, the vehicle display device **1** conducts a display in such a way that the caution image **50** is superimposed onto the motorcycle that is about to pass on the side of the vehicle **100**. The caution image **50** is displayed in/with a color or brightness that easily attracts attention of the driver **200**. Therefore, the driver **200** can grasp a movement of the motorcycle while gazing forward without directing a line of sight to the peripheral image **40**.

[0058] As described above, the vehicle display device **1** according to the present embodiment includes the image acquisition unit **31** and the display device **2**. The image acquisition unit **31** acquires the peripheral image **40**. The peripheral image **40** is an image obtained by imaging a peripheral region of the vehicle **100** including a rear region of the vehicle **100**. The display device **2** displays the peripheral image **40** in the display region **120** in the windshield **110** of the vehicle **100**.

[0059] The display region **120** is an upper region in the windshield **110**, and is a region that is

continuous from one end to another end in the vehicle width direction Y in the windshield **110**. The vehicle display device **1** according to the present embodiment displays the peripheral image **40** in a region that is wide in the vehicle width direction Y, and this can reduce a movement of a line of sight of the driver **200** that is required to check the periphery.

[0060] The display device **2** according to the present embodiment includes a transparent display that is disposed in the windshield **110**. The peripheral image **40** is displayed by this display. This enables the peripheral image **40** to be displayed without obstructing a front field of view of the driver **200**.

[0061] The peripheral image **40** according to the present embodiment is a continuous image obtained by imaging a region from a diagonally left rear side to a diagonally right rear side of the vehicle **100**. An image indicating a wide range, as described above, is displayed, and this effectively reduces a movement of a line of sight at a time when the driver **200** checks the periphery of the vehicle **100**.

[0062] The peripheral image **40** according to the present embodiment is an image obtained by imaging a region of a rear viewing angle θ serving as an angle of a field of view in the horizontal direction in a case where a rear side is viewed from the predetermined point P1 of the vehicle **100**. An angle of the rear viewing angle θ is appropriately set, and this can provide the driver **200** with an appropriate rear field of view.

[0063] The peripheral image **40** may be a continuous image obtained by imaging a region from a left-hand side to a right-hand side of the vehicle **100**. Such a peripheral image **40** is generated from, for example, images captured by a plurality of cameras **130**. The peripheral image **40** includes images indicating sides of the vehicle **100**, and this effectively reduces a movement of a line of sight at a time when the driver **200** checks the periphery of the vehicle **100**.

[0064] The vehicle display device **1** according to the present embodiment includes the generation unit **34**. The generation unit **34** generates the caution image **50** serving as an image for cautioning against the target to be careful about that is included in the peripheral image **40**. The display device **2** displays the peripheral image **40** and the caution image **50** in the display region **120**. The vehicle display device **1** according to the present embodiment displays the caution image **50**, and this can call attention of the driver **200** to the target to be careful about.

[0065] An image display method according to the present embodiment includes a step of displaying the peripheral image **40** in the display region **120** in the windshield **110** of the vehicle **100**. In the flowchart of FIG. **8**, the peripheral image **40** is displayed in the display region **120** in Step S60, Step S90, and Step S100. The image display method according to the present embodiment can reduce a movement of a line of sight of the driver **200** that is required to check the periphery.

[0066] Note that the position and the shape that are illustrated of the display region **120** are examples, and are not limited to this position and this shape. For example, a length in the vehicle width direction Y of the display region **120** may be determined according to an angle of view of the peripheral image **40** to be displayed. In a case where an enough length to display the peripheral image **40** has been secured, there may be a clearance between the display region **120** and the vehicle body **170**.

Variation of Embodiment

[0067] A variation of the embodiment will be described with reference to FIG. **11**. FIG. **11** is a diagram illustrating a schematic configuration of a display device according to the variation of the embodiment. A vehicle display device **1** according to the variation of the embodiment is different from the vehicle display device **1** according to the embodiment described above, for example, in that an image is projected onto the display region **120**.

[0068] As illustrated in FIG. **11**, a display device **2** according to the variation of the embodiment is what is called a head-up display device. The display device **2** includes a display unit **20**, a backlight unit **21**, a mirror **22**, a housing **23**, and a transparent cover **24**. The display unit **20**, the backlight unit **21**, and the mirror **22** are housed in the housing **23**. The cover **24** closes an opening of the

housing **23**. The opening of the housing **23** faces the windshield **110** in a vehicle upward/downward direction Z.

[0069] The display unit **20** is, for example, a liquid crystal display device such as a thin film transistor-liquid crystal display (TFT-LCD). An image of the display unit **20** is output as display light Lt by using light of the backlight unit **21**. The mirror **22** reflects the display light Lt toward the display region **120** of the windshield **110**. The display light Lt that has been transmitted through the cover **24** is reflected toward the driver **200** by the display region **120**. The driver **200** visually recognizes a virtual image Vi by using the display light Lt.

[0070] In the vehicle display device **1** illustrated in FIG. **11**, a control unit **3** is housed in the housing **23**. The control unit **3** acquires the peripheral image **40** to generate the caution image **50** similarly to the control unit **3** according to the embodiment described above. The control unit **3** causes the display unit **20** to display the peripheral image **40** and the caution image **50**. This causes the peripheral image **40** and the caution image **50** to be displayed in the display region **120** of the windshield **110**.

[0071] The virtual image Vi including the peripheral image **40** and the caution image **50** is formed in a position before the windshield **110**. Accordingly, a movement of a line of sight is reduced in a case where the driver **200** visually recognizes the peripheral image **40** and the caution image **50**, and an amount of adjustment of a focal length is also reduced.

[0072] Note that the vehicle display device **1** can adjust a position of projection of the display light Lt in accordance with a detection result of a driver monitor **7**. The driver monitor **7** is a camera that images the driver **200**. An eye point EP of the driver **200** is detected on the basis of an image captured by the driver monitor **7**. The control unit **3** adjusts a direction of projection of the display light Lt in accordance with the detected eye point EP. The adjustment of the direction of projection is achieved, for example, by rotationally moving the mirror **22**.

[0073] Note that the vehicle display device **1** may include, as the display device **2**, the head-up display device according to the variation in addition to the transparent display according to the embodiment. In this case, the peripheral image **40** may be displayed by the transparent display, and the caution image **50** may be displayed by the head-up display device.

[0074] Pieces of content disclosed in the embodiment and the variation that have been described above can be appropriately combined and implemented.

REFERENCE SIGNS LIST

[0075] In the vehicle display device according to the present embodiment, the display region of the windshield is an upper region in the windshield, and is a region that is continuous from one end to another end in the vehicle width direction in the windshield. The vehicle display device according to the present embodiment displays, in the display region, an image indicating a wide range of the rear side of the vehicle, and this exhibits an advantageous effect in which an amount of movement of a line of sight of a driver can be reduced in a case where the driver checks the periphery of the vehicle.

[0076] Although the invention has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

Claims

1. A vehicle display device comprising: an image acquisition unit that acquires a peripheral image being an image obtained by imaging a peripheral region of a vehicle, the peripheral region including a rear region of the vehicle; a display device that displays the peripheral image in a display region in a windshield of the vehicle; and a generation unit that generates a caution image being an image for cautioning against a target to be careful about that is included in the peripheral

image, wherein the display region is an upper region in the windshield, and is a region that is continuous from one end to another end in a vehicle width direction in the windshield, the target to be careful about at the time of backward movement of the vehicle is an object that is present on a route behind the vehicle or an object that is present near the route, the target to be careful about at the time of a forward movement of the vehicle is an object that is approaching to the vehicle from a rear side, and the display device displays the peripheral image and the caution image in the display region.

2. The vehicle display device according to claim 1, wherein the display device includes a display that is transparent and is disposed in the windshield, and the peripheral image is displayed by the display.

3. The vehicle display device according to claim 1, wherein the peripheral image is a continuous image obtained by imaging a region from a diagonally left rear side to a diagonally right rear side of the vehicle.

4. The vehicle display device according to claim 3, wherein the peripheral image is an image obtained by imaging a region of a rear viewing angle being an angle of a field of view in a horizontal direction in a case where a rear side is viewed from a predetermined point of the vehicle.

5. The vehicle display device according to claim 1, wherein the peripheral image is a continuous image obtained by imaging a region from a left-hand side to a right-hand side of the vehicle.

6. An image display method comprising: a step of displaying a peripheral image and a caution image being an image for cautioning against a target to be careful about that is included in the peripheral image in a display region in a windshield of a vehicle, wherein the peripheral image is an image obtained by imaging a peripheral region of the vehicle, the peripheral region including a rear region of the vehicle, the display region is an upper region in the windshield, and is a region that is continuous from one end to another end in a vehicle width direction in the windshield, the target to be careful about at the time of backward movement of the vehicle is an object that is present on a route behind the vehicle or an object that is present near the route, and the target to be careful about at the time of a forward movement of the vehicle is an object that is approaching to the vehicle from a rear side.
